ChE 312 – Transport Phenomena II –Spring 2020
Student Learning Outcome 1
Rubric for assessment problem on final exam (Problem #2)
Heat Transfer .

SLO 1 Performance Indicator (PI)	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem (the same for all items) 40 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<32 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (32-37 pts)	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (>37 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem (the same for all items) 40 pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<32 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (32-37 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (>37 pts)
4. Translates academic theory into engineering applications Problem (the same for all items) 40 pts	Does not appear to grasp the connection between theory and the problem (<32 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (32-37 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (>37 pts)

Averages:

Performance Indicators: (1) 2.57   Overall Average: 2.57

Average score was fairly high for this problem. Clearly, most students were strong in formulating the engineering system and model equations, and a majority 47/49 were satisfactory or higher in applying the engineering principles to solve this problem.

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QUESTION 2: (40 pts)

Water enters a tube at 27 °C with a flow rate of 450 kg/h. The heat transfer from the tube wall to the fluid is given as q' (W/m) = $\alpha \cdot x$, where the coefficient α is 20 W/m^2 and x(m) is the axial distance from the tube entrance. Beginning with a properly defined differential control volume in the tube, derive an expression for the temperature distribution $T_m(x)$ of the water. What is the outlet temperature of the water for a heated section 30 m long? (Hint: for water, C_P = 4.18 kJ/(kg·K).

Solution

Solution set

CHE 201 Final Exam May 10th, 2018

QUESTION 4: (20 pts)

A piston-cylinder device which contains 3 kg of saturated water vapor at 6 bar. Heat is transferred between the piston-cylinder and a 25°C thermal reservoir until the saturated water vapor becomes a saturated liquid at 6 bar. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored. Steam Tables are at the end of Exam.

+2 a) (2 pts) Draw a schematic of the system and label the system boundary.

+4 b) (4 pts) State the engineering model

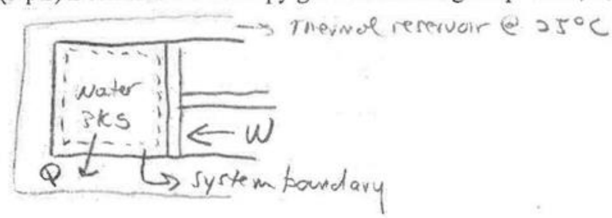
+4 c) (4 pts) Write the simplified energy balance and simplified entropy balance.

d) (10 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.

e) (5 pts) Determine the change in entropy of the water during the process, **ΔS, in kJ/K**

f) (5 pts) Determine the entropy generated during the process, **σ, in kJ/K.**

a)



b) 1) Water in a piston-cylinder assembly is the closed system

2) ΔKE + ΔPE are ignored

3) Heat transfer to thermal reservoir

4) Volume change is the work mode

c) Energy balance: $\Delta U + \cancel{\Delta KE} + \cancel{\Delta PE} = Q - W$

$$\Delta U = Q - W = m(u_2 - u_1) = Q - W$$

Entropy Balance: $\Delta S = \int_1^2 \left(\frac{\delta Q}{T}\right)_b + \sigma_{cv}$

d) Work (kJ) $= \int_1^2 P\,dv$ SAT vapor → SAT Liquid
Constant P = 6 bar

Work (kJ) $= mP(v_2 - v_1)$ Table A-3

$v_2 = 1.1006\times10^{-3}\ m^3/kg$
$v_1 = 0.3157\ m^3/kg$
$m = 3\ kg$

Work (kJ) $= (3\ kg)(6\ bar)(1.1006\times10^{-3} - 0.3157)\frac{m^3}{kg}$

$= (18)(-0.3146)\ bar\text{-}m^3$

$= -5.66\ bar\text{-}m^3 \left|\frac{10^5\ N/m^2}{1\ bar}\right|\frac{1\ kJ}{10^3\ N\text{-}m} = \boxed{-566.3\ kJ = Work}$

on system

Heat (kJ): $Q = m(u_2 - u_1) + W$

$u_2 = 669.90\ kJ/kg$
$u_1 = 2567.4$ "

$Q(kJ) = (3\ kg)(669.9 - 2567.4)\frac{kJ}{kg} + (-566.3\ kJ)$

$Q = (3(-1897.5) - 566.3)\ kJ$

$Q = -5692.5 - 566.3 = \boxed{-6258.8\ kJ = Q}$

From system

e) $\Delta S = ?$ $s_1 = 6.76\ \frac{kJ}{kg\text{-}K}$
$s_2 = 1.9312$ "

$\Delta S = (3\ kg)(1.9312 - 6.760)\frac{kJ}{kg\text{-}K}$

$= (3)(-4.8288)\frac{kJ}{K}$

$\boxed{\Delta S = -14.4864\ kJ/K}$

f) $\sigma_{cv} = \Delta S - \left(\frac{Q}{T}\right)_b$ $T_b = 25°C + 273.15\ K = 298.15\ K$

$\sigma_{cv} = -14.4864\frac{kJ}{K} - \left(\frac{-6258.8\ kJ}{298.15\ K}\right) =$

$= -14.4864\frac{kJ}{K} + 20.992\frac{kJ}{K} = \boxed{6.5056\frac{kJ}{K} = \sigma_{cv}}$

$\sigma_{cv} > 0$ possible!